

Shallow vs Deep Networks

A Shallow Neural Network typically consists of one or two hidden layers, whereas a Deep Neural Network (DNN) has multiple hidden layers (usually more than three).

Feature	Shallow	Deep
# of layers	1-2	3>
Feature extraction	manual	Network extracts the features
Training time	Faster to train	Requires more time and computational power
Data size	Trains well on small datasets	Performs better with larger datasets
Complexity	Can't handle high-dimensional data	Excels in recognizing complex patterns
Overfitting	Lower risk	Requires regularization techniques

Deciding on which to use depends mainly on the problem, but as a general rule of thumb DNNs work better on complex problems where complex patterns exist, DNNs excel in extracting these patterns, but if the problem is simple enough shallow networks might be able to solve it.

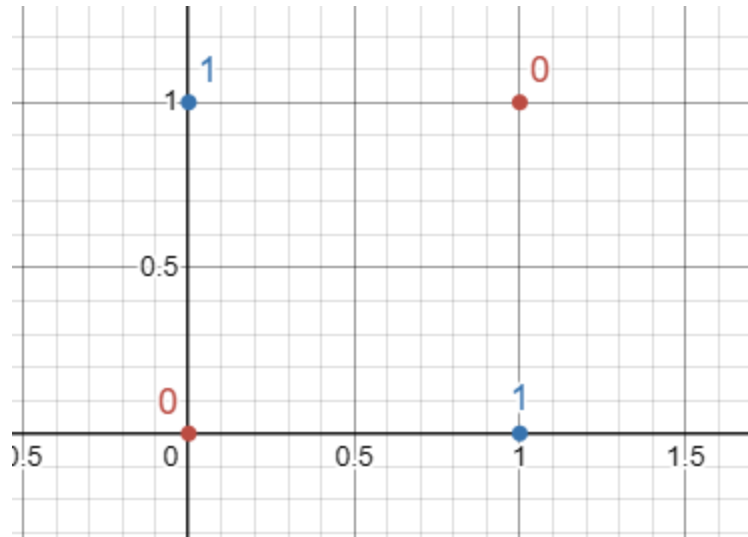
As a demonstration on how limited shallow networks are we will discuss the XOR problem: The XOR (Exclusive OR) problem is a classic problem in neural networks and deep learning. It demonstrates why single-layer perceptrons (simple shallow networks) fail and why multi-layer perceptrons (MLPs) with hidden layers are necessary.

The XOR function is a binary classification task where:

- Input: Two binary values (0 or 1).
- Output: 1 if inputs are different, 0 if they are the same.

A single-layer perceptron can only solve problems that are linearly separable (i.e., can be separated by a straight line).

However, XOR is not linearly separable as shown in the following graph, we can't draw a single straight line that solves the problem.



To solve XOR, we need at least one hidden layer with non-linear activation functions (like ReLU or sigmoid), in a nutshell we can introduce non-linearity to the system by using more hidden layers and using suitable activation functions.